

Week 6 – Magnetism: Anomalies and Susceptibility

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Workshop Learning Goals

By the end of this workshop, students should be able to:

- Visualise magnetic fields generated by dipoles and combinations of sources
- Relate magnetic susceptibility to rock type and geological process
- Predict magnetic anomaly shapes from simple geological geometries
- Contrast magnetic anomalies with gravity anomalies for the same structures
- Explain the concept of Curie depth and its relationship to thermal structure
- Link source depth to anomaly wavelength and spectral content

Timeline and Flow

0–10 min Introduction and framing

- How magnetism differs from gravity (vector nature, polarity, temperature dependence)
- Overview of the three activities and how they connect

10–35 min Activity 1: Magnetic fields from dipoles

- Iron filings and disc magnets
- Vertical vs horizontal dipole orientation
- Stacked magnets and multiple sources

Learning emphasis:

- Dipole field geometry
- Orientation-dependent symmetry and polarity
- Intuition for induced magnetisation

35–55 min Whole-class demo: Magnetic susceptibility of rocks

- Susceptibility measurements of igneous, metamorphic, and sedimentary rocks
- Effects of alteration, metamorphism, and weathering

Learning emphasis:

- Susceptibility as a material property
- Why similar compositions can have very different magnetic behaviour

55–95 min Activity 2: Magnetic anomalies vs gravity

- Reasoning-based sketching of anomalies for idealised geometries
- Direct comparison with gravity anomalies

95–105 min Break**105–160 min Activity 3: Curie depth and magnetic crust**

- Curie temperature and magnetic crustal thickness
- Depth–wavelength relationships
- Antarctica case study
- Back-of-the-envelope heat-flow estimates

160–180 min Synthesis and wrap-up

- What magnetism can and cannot tell us
- Connection to upcoming Fourier/spectral methods